

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 11: Finding Common Ground

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	11 — Finding Common Ground
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Each problem may chain several ideas: decide HCF (largest divider — share/cut/tile, or 'same unknown remainder' via the HCF of pairwise differences) versus LCM (smallest common multiple — meet/ring/flash again, or $\text{LCM} \cdot k + r$ for a fixed remainder); always take the LOWEST powers of COMMON primes for HCF and the HIGHEST powers of ALL primes for LCM; recall $\text{HCF} \cdot \text{LCM} = \text{product}$ holds for exactly two numbers while $\text{LCM} = \text{product}$ holds only for co-primes; count the co-prime factor-splits of LCM/HCF when asked how many pairs share a given HCF and LCM; and watch unit conversions, 'including/excluding the start' tallies, and which runner is fastest versus slowest. Do not write on this paper — mark answers on your answer sheet.

1. Two numbers factorise as $1080 = 2^3 \times 3^3 \times 5$ and $2100 = 2^2 \times 3 \times 5^2 \times 7$. The ratio LCM : HCF for this pair equals:

- (A) 210 (B) 60
(C) 37800 (D) 630

2. How many factors of 1800 are perfect squares (including 1)?

- (A) 8 (B) 6
(C) 9 (D) 4

3. The total number of factors of 7560 is:

- (A) 64 (B) 48
(C) 32 (D) 60

4. How many factors of 1800 are even?

- (A) 9 (B) 36
(C) 27 (D) 18

5. The smallest natural number that has exactly 18 factors and is a multiple of 24 is:

- (A) 144 (B) 240
(C) 360 (D) 288

6. A number $N = 2^a \times 3^b$ has exactly 12 factors with $a > b \geq 1$, and N is as small as possible. The value of N is:

- (A) 96 (B) 144
(C) 72 (D) 108

7. The largest 3-digit number of the form $2^a \times 3^b \times 5^c$ with a, b, c each at least 1 is:

- (A) 960 (B) 990
(C) 900 (D) 720

8. The numbers $2^4 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 5^2 \times 7$ are given. Their HCF is:

- (A) 60 (B) 180
(C) 900 (D) 2520

9. For the numbers $2^4 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 5^2 \times 7$, the LCM is:

- (A) $2^4 \times 3^3 \times 5^2 \times 7$ (B) $2^2 \times 3^2 \times 5$
(C) $2^6 \times 3^5 \times 5^3 \times 7$ (D) $2^4 \times 3^3 \times 5^2$

10. The number of common factors of 540 and 756 is:

- (A) 9 (B) 6
(C) 12 (D) 18

11. Two numbers in the ratio 9 : 14 have an HCF of 8. Their LCM is:

- (A) 112 (B) 1008
(C) 126 (D) 896

12. Two numbers in the ratio 7 : 12 have an LCM of 420. Their HCF is:

- (A) 5 (B) 7
(C) 84 (D) 12

13. The product of two numbers is 4320 and their HCF is 12. Their LCM is:

- (A) 51840 (B) 30
(C) 360 (D) 120

14. The product of two numbers equals 18 times the square of their HCF. If their HCF is 15, their LCM is:

- (A) 4050 (B) 225
(C) 18 (D) 270

15. Two numbers have HCF 18 and LCM 1260. If one number is 126, the other is:

- (A) 140 (B) 1260
(C) 162 (D) 180

16. For two numbers, $\text{LCM} = 21 \times \text{HCF}$ and the HCF is 12. The product of the two numbers is:

- (A) 252 (B) 144
(C) 63504 (D) 3024

17. The product of two co-prime numbers, each greater than 1, is 1763. The two numbers are:

- (A) 1 and 1763 (B) 41 and 43
(C) 23 and 77 (D) 37 and 49

18. The HCF of two numbers is 12 and their LCM is 720. The number of unordered pairs (a, b) with $a < b$ that have this HCF and LCM is:

- (A) 3 (B) 2
(C) 1 (D) 4

19. The HCF of two numbers is 9 and their LCM is 1890. How many unordered pairs (a, b) with $a < b$ have this HCF and LCM?

- (A) 4 (B) 8
(C) 16 (D) 6

20. Two numbers have HCF 7 and their LCM is $7 \times 11 \times 13 \times 17 \times 19$. How many unordered pairs (a, b) with $a < b$ are possible?

- (A) 8 (B) 16
(C) 4 (D) 5

21. The sum of two numbers is 1080 and their HCF is 36. How many unordered pairs of such numbers are possible?

- (A) 3 (B) 4
(C) 5 (D) 6

22. A number n satisfies $\text{HCF}(n, 48) = 16$ and $\text{LCM}(n, 48) = 240$. The number n is:

- (A) 240 (B) 16
(C) 120 (D) 80

23. Which of these CANNOT be the HCF and LCM of some pair of numbers?

- (A) HCF 18, LCM 250 (B) HCF 16, LCM 240
(C) HCF 14, LCM 280 (D) HCF 15, LCM 450

24. A florist has 480 roses, 600 lilies and 720 tulips. She makes identical bouquets using all the flowers, each bouquet having the same number of each kind. The largest number of bouquets is:

- (A) 240
(C) 120
- (B) 60
(D) 24

25. For the same florist (480 roses, 600 lilies, 720 tulips, 120 bouquets), the number of lilies in each bouquet is:

- (A) 4
(C) 5
- (B) 6
(D) 3

26. Square tiles of the largest possible whole-cm side exactly pave a room 11 m 34 cm long and 7 m 56 cm wide. The side of each tile is:

- (A) 189 cm
(C) 378 cm
- (B) 126 cm
(D) 54 cm

27. For the room of the previous setting (1134 cm by 756 cm, tile side 378 cm), the least number of square tiles required is:

- (A) 6
(C) 3
- (B) 12
(D) 8

28. Three ropes of lengths 8 m, 14 m and 21 m are cut into equal pieces of the greatest possible length with nothing wasted. The length of each piece (in cm) is:

- (A) 100 cm
(C) 200 cm
- (B) 50 cm
(D) 700 cm

29. Beads numbering 493 red, 551 blue and 667 green are strung so that each string holds one colour only and every string has the same largest number of beads. That number is:

- (A) 29
(C) 19
- (B) 17
(D) 23

30. A merchant has 221 L, 323 L and 391 L of three oils. The largest can that can measure each an exact whole number of times has capacity:

- (A) 13 L
(C) 17 L
- (B) 19 L
(D) 23 L

31. A field 1517 m by 1739 m is divided into identical square plots of the greatest possible size. The number of plots along the 1739 m side is:

- (A) 41
(C) 37
- (B) 47
(D) 53

32. A gardener has 1056 marigolds and 792 roses, arranging each kind into rows of equal, largest possible size (one kind per row). The total number of rows is:

- (A) 14
(C) 21
- (B) 12
(D) 7

33. Coloured cards numbering 825 red, 675 blue and 1125 green are bundled so each bundle holds one colour only and all bundles are equal and as large as possible. Each

bundle contains:

- | | |
|---------|--------|
| (A) 75 | (B) 25 |
| (C) 150 | (D) 15 |

34. A shopkeeper has 144 kg of rice, 180 kg of wheat and 240 kg of pulses, packing each grain into bags of equal, largest possible size (one grain per bag). The total number of bags is:

- | | |
|--------|--------|
| (A) 12 | (B) 47 |
| (C) 94 | (D) 20 |

35. The greatest length of a measuring rod that can measure exactly each of 11 m 5 cm, 15 m 47 cm and 19 m 89 cm is:

- | | |
|------------|-------------|
| (A) 13 cm | (B) 17 cm |
| (C) 221 cm | (D) 1105 cm |

36. Four bells ring at intervals of 24, 36, 48 and 60 minutes. Ringing together now, they next ring together after:

- | | |
|-----------------|------------------|
| (A) 360 minutes | (B) 1440 minutes |
| (C) 240 minutes | (D) 720 minutes |

37. Three runners complete a lap in 63 s, 84 s and 105 s. Starting together, the number of complete laps the FASTEST runner finishes by the time they next all meet is:

- | | |
|--------|--------|
| (A) 15 | (B) 12 |
| (C) 20 | (D) 16 |

38. Cyclists lap a track every 40 s, 48 s and 60 s. Starting together, the number of complete laps the SLOWEST cyclist has finished when all three next meet is:

- | | |
|-------|--------|
| (A) 6 | (B) 4 |
| (C) 5 | (D) 10 |

39. Two gears mesh, one with 40 teeth and the other with 48 teeth. After how many teeth of the SMALLER gear pass do they return to their starting alignment?

- | | |
|----------------|---------------|
| (A) 1920 teeth | (B) 8 teeth |
| (C) 120 teeth | (D) 240 teeth |

40. Bells P, Q and R toll every 6, 9 and 12 seconds, starting together. In 30 minutes, how many times do all three toll together, NOT counting the toll at the very start?

- | | |
|--------|--------|
| (A) 51 | (B) 25 |
| (C) 49 | (D) 50 |

41. Four lights flash every 10, 15, 20 and 25 seconds, starting together. In one hour, how many times do all four flash together, INCLUDING the flash at the start?

- | | |
|--------|--------|
| (A) 12 | (B) 13 |
| (C) 20 | (D) 61 |

42. Three signals blink every 14, 21 and 35 minutes. They blink together at 9:00 am. The next time all three blink together is:

- | | |
|--------------|--------------|
| (A) 11:00 am | (B) 12:30 pm |
| (C) 9:42 am | (D) 3:30 pm |

43. Three traffic lights change every 42 s, 70 s and 84 s. Having just changed together, after how long do all three next change together?

- | | |
|-----------------|-----------------|
| (A) 840 seconds | (B) 210 seconds |
| (C) 14 seconds | (D) 420 seconds |

44. Two buses leave a stop every 45 minutes and every 60 minutes, together at 6:00 am. They next leave together at:

- | | |
|-------------|----------------|
| (A) 7:30 am | (B) 8:15 am |
| (C) 9:00 am | (D) 12:00 noon |

45. On a circular track, A laps every 56 s, B every 70 s and C every 84 s, all starting together. By the time they first all return to the start together, how many MORE laps has A completed than C?

- | | |
|--------|-------|
| (A) 10 | (B) 5 |
| (C) 15 | (D) 3 |

46. Three clocks chime every 30, 45 and 75 seconds, starting together at exactly 10:00:00. The first time after 10:00 that all three chime together again is:

- | | |
|--------------|--------------|
| (A) 10:04:30 | (B) 10:07:30 |
| (C) 10:15:00 | (D) 10:01:15 |

47. The smallest number which, when divided by 12, 15 and 20, leaves a remainder of 7 in each case is:

- | | |
|--------|---------|
| (A) 60 | (B) 127 |
| (C) 47 | (D) 67 |

48. The greatest 3-digit number that leaves remainder 5 when divided by each of 8, 12 and 18 is:

- | | |
|---------|---------|
| (A) 936 | (B) 941 |
| (C) 933 | (D) 965 |

49. The smallest number which, when increased by 13, becomes exactly divisible by 18, 24 and 30 is:

- | | |
|---------|---------|
| (A) 360 | (B) 373 |
| (C) 347 | (D) 13 |

50. The largest number that divides 2961, 5097 and 6873 leaving the same remainder in each case is:

- | | |
|--------|--------|
| (A) 24 | (B) 12 |
| (C) 48 | (D) 8 |

51. A number leaves remainder 1 when divided by each of 2, 3, 4, 5 and 6, and leaves remainder 0 when divided by 7. The smallest such number greater than 1 is:

- | | |
|---------|---------|
| (A) 121 | (B) 301 |
| (C) 421 | (D) 61 |

52. A number when divided by 12 leaves remainder 5, and when divided by 18 leaves remainder 11. The smallest such positive number is:

- | | |
|--------|--------|
| (A) 41 | (B) 5 |
| (C) 36 | (D) 29 |

53. The smallest number greater than 2 that leaves remainder 2 when divided by each of 5, 6 and 8, and is itself a multiple of 7 is:

- | | |
|---------|---------|
| (A) 602 | (B) 122 |
| (C) 482 | (D) 722 |

54. The smallest number which, when divided by 16, 24 and 40, leaves remainders 9, 17 and 33 respectively, is:

- | | |
|---------|---------|
| (A) 240 | (B) 233 |
| (C) 247 | (D) 113 |

55. Find the smallest 4-digit number that is exactly divisible by 8, 12 and 18.

- | | |
|----------|----------|
| (A) 1000 | (B) 1080 |
| (C) 1008 | (D) 1044 |

56. A number leaves remainder 4 when divided by 6, remainder 4 when divided by 9, and remainder 4 when divided by 15. The smallest such number greater than 4 is:

- | | |
|---------|--------|
| (A) 94 | (B) 90 |
| (C) 184 | (D) 49 |

57. Two co-prime numbers, each greater than 1, have a product of 1591. The two numbers are:

- | | |
|----------------|---------------|
| (A) 1 and 1591 | (B) 31 and 51 |
| (C) 41 and 39 | (D) 37 and 43 |

58. Three numbers a, b, c are pairwise co-prime (every pair has HCF 1). Which set has this property?

- | | |
|------------------|-------------------|
| (A) 9, 16 and 35 | (B) 8, 12 and 35 |
| (C) 6, 25 and 21 | (D) 10, 21 and 14 |

59. A whole number is divisible by both 18 and by 24 but is NOT necessarily divisible by 16. It is always divisible by:

- | | |
|---------|---------|
| (A) 432 | (B) 144 |
| (C) 72 | (D) 36 |

60. Among the whole numbers from 1 to 600, how many are common multiples of 16 and 18?

(A) 8

(B) 37

(C) 4

(D) 33

Solutions — Mathematics (Class 7), Chapter 11: Finding Common Ground — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	B	31	B	41	B	51	B
2	A	12	A	22	D	32	D	42	B	52	D
3	A	13	C	23	A	33	A	43	D	53	A
4	C	14	D	24	C	34	B	44	C	54	B
5	D	15	D	25	C	35	C	45	B	55	C
6	C	16	D	26	C	36	D	46	B	56	A
7	A	17	B	27	A	37	C	47	D	57	D
8	B	18	D	28	A	38	B	48	B	58	A
9	A	19	B	29	A	39	D	49	C	59	C
10	C	20	A	30	C	40	D	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) The ratio LCM : HCF equals the product of each prime raised to the difference of its two exponents. For 2: $3 - 2 = 1$; for 3: $3 - 1 = 2$; for 5: $2 - 1 = 1$; for 7: $1 - 0 = 1$. So $\text{LCM}/\text{HCF} = 2^1 \times 3^2 \times 5^1 \times 7^1 = 2 \times 9 \times 5 \times 7 = 630$. The value 60 is the HCF itself, 37800 is the LCM itself, and 210 drops the square on the 3.

2. (A) $1800 = 2^3 \times 3^2 \times 5^2$. A square factor uses only even exponents. For 2 the even choices are 0 or 2 (2 ways), for 3 they are 0 or 2 (2 ways), for 5 they are 0 or 2 (2 ways). That gives $2 \times 2 \times 2 = 8$ square factors. Counting odd exponents too gives the full factor count, and miscounting the 2-prime as having three even options leads to 9.

3. (A) $7560 = 2^3 \times 3^3 \times 5 \times 7$. The number of factors is $(3+1)(3+1)(1+1)(1+1) = 4 \times 4 \times 2 \times 2 = 64$. Forgetting the +1 on one prime, or multiplying exponents instead of (exponent+1), produces the smaller wrong totals.

4. (C) $1800 = 2^3 \times 3^2 \times 5^2$ has $(3+1)(2+1)(2+1) = 36$ factors in all. The ODD factors use 2^0 only: $(2+1)(2+1) = 9$ of them. Even factors = total - odd = $36 - 9 = 27$. Mistaking 9

(the odd count) for the answer is the trap.

5. (D) A multiple of 24 must contain $2^3 \times 3$ at least. To get exactly $18 = 2 \times 3 \times 3$ factors with small value, try $2^5 \times 3^2 = 32 \times 9 = 288$, which has $(5+1)(2+1) = 18$ factors and is divisible by 24. Smaller multiples of 24 such as $144 = 2^4 \times 3^2$ give $(4+1)(2+1) = 15$ factors, and 240 gives 20 factors, so 288 is the least.

6. (C) Exactly 12 factors means $(a+1)(b+1) = 12$ with $a > b \geq 1$. The valid factorisations are $a=3, b=2$ giving $N = 2^3 \times 3^2 = 72$, and $a=5, b=1$ giving $N = 2^5 \times 3 = 96$. The smallest is 72. The value 96 is the larger valid case, $144 = 2^4 \times 3^2$ has 15 factors, and $108 = 2^2 \times 3^3$ has b greater than a .

7. (A) We need a 3-digit number divisible by $2 \times 3 \times 5 = 30$, of the form $2^a 3^b 5^c$ with no other primes. Among multiples of 30 below 1000, $960 = 2^6 \times 3 \times 5$ qualifies (only primes 2,3,5). The larger $990 = 2 \times 3^2 \times 5 \times 11$ contains the prime 11, so it is disqualified; $900 = 2^2 \times 3^2 \times 5^2$ is valid but smaller than 960.

8. (B) HCF uses the lowest power of each COMMON prime: $2 \rightarrow \min(4,2) = 2^2$; $3 \rightarrow \min(2,3) = 3^2$; $5 \rightarrow \min(1,2) = 5^1$; the prime 7 appears in only one number so it is excluded. $\text{HCF} = 4 \times 9 \times 5 = 180$. Taking highest powers gives a much larger LCM, and including 7 or mis-taking 5^2 inflates the answer.

9. (A) LCM uses the highest power of EVERY prime present: $2 \rightarrow \max(4,2) = 2^4$; $3 \rightarrow \max(2,3) = 3^3$; $5 \rightarrow \max(1,2) = 5^2$; $7 \rightarrow 7^1$. So $\text{LCM} = 2^4 \times 3^3 \times 5^2 \times 7$. Using lowest powers gives the HCF, ADDING exponents ($2^6 \times 3^5 \dots$) is the classic error, and dropping the 7 leaves out a prime that is genuinely present.

10. (C) Common factors of two numbers are exactly the factors of their HCF. $\text{HCF}(540, 756) = 108 = 2^2 \times 3^3$, which has $(2+1)(3+1) = 12$ factors. So there are 12 common factors. Using the wrong HCF, or counting factors of $36 = 2^2 \times 3^2$ (which gives 9), is the trap.

11. (B) Since 9 and 14 are co-prime, the numbers are $9 \times 8 = 72$ and $14 \times 8 = 112$, and $\text{HCF} = 8$ as required. $\text{LCM} = 9 \times 14 \times 8 = 1008$ (equivalently the product $72 \times 112 = 8064$ divided by $\text{HCF } 8$). The value 112 is just the larger number, and $126 = 9 \times 14$ forgets the common factor.

12. (A) With co-prime ratio parts 7 and 12, the numbers are $7h$ and $12h$, and their LCM is $7 \times 12 \times h = 84h$. Setting $84h = 420$ gives $h = 5$, the HCF. The numbers are 35 and 60. Confusing the HCF with a ratio part (7 or 12), or with $84 = 7 \times 12$, are the traps.

13. (C) For two numbers, $\text{HCF} \times \text{LCM} = \text{product}$. So $\text{LCM} = \text{product} / \text{HCF} = 4320 / 12 = 360$. Multiplying instead (4320×12) or dividing the wrong way ($12/4320$ then inverted to give 30 or 120) are the errors.

14. (D) Product = $18 \times (\text{HCF})^2 = 18 \times 225 = 4050$. Since $\text{HCF} \times \text{LCM} = \text{product}$, $\text{LCM} = \text{product} / \text{HCF} = 4050 / 15 = 270$. Equivalently, dividing product = $18 \times \text{HCF}^2$ by HCF gives $\text{LCM} = 18 \times \text{HCF} = 18 \times 15 = 270$. Reporting the product 4050 or $\text{HCF}^2 = 225$ instead of the LCM is the trap.

15. (D) Product of the two numbers = $\text{HCF} \times \text{LCM} = 18 \times 1260 = 22680$. The other number = $22680 / 126 = 180$. Check: $\text{HCF}(126, 180) = 18$ and $\text{LCM}(126, 180) = 1260$. The value $140 = \text{LCM}/9$ is a miscalculation, and 1260 just repeats the LCM.

16. (D) $\text{LCM} = 21 \times 12 = 252$. Product = $\text{HCF} \times \text{LCM} = 12 \times 252 = 3024$. The value 252 is only the LCM, $144 = \text{HCF}^2$, and $63504 = 252^2$ squares the LCM by mistake.

17. (B) Co-prime numbers both above 1 with product 1763 must be its two prime factors. $1763 = 41 \times 43$, and 41, 43 are both prime so $\text{HCF} = 1$. The pair 1 and 1763 fails 'each greater than 1'; $23 \times 77 = 1771$ not 1763; and $37 \times 49 = 1813$ not 1763.

18. (D) Write $a = 12m$, $b = 12n$ with $\text{HCF}(m, n) = 1$ and $m \times n = 720/12 = 60 = 2^2 \times 3 \times 5$. The co-prime factor pairs of 60 are (1, 60), (4, 15), (3, 20), (5, 12) — four unordered pairs (split each of the 3 prime-power blocks to one side: $2^{(3-1)} = 4$ ways). So 4 pairs exist.

19. (B) Set $a = 9m$, $b = 9n$, $\text{HCF}(m, n) = 1$, $m \times n = 1890/9 = 210 = 2 \times 3 \times 5 \times 7$. With four distinct prime blocks, the number of co-prime ordered splits is $2^4 = 16$, so unordered pairs = $16/2 = 8$. The value 4 corresponds to only three prime blocks.

20. (A) Removing the common factor 7, the remaining product $11 \times 13 \times 17 \times 19$ has 4 distinct prime blocks to distribute between the two co-prime parts: $2^4 = 16$ ordered, hence $16/2 = 8$ unordered pairs with $a < b$. Forgetting to halve gives 16, and using three blocks gives 4.

21. (B) Let the numbers be $36m$ and $36n$ with $\text{HCF}(m, n) = 1$. Then $36(m + n) = 1080$, so $m + n = 30$. Co-prime pairs with $m < n$ summing to 30: (1, 29), (7, 23), (11, 19), (13, 17) — four pairs. Pairs like (3, 27) or (5, 25) share a factor and are excluded.

22. (D) Using $\text{HCF} \times \text{LCM} = n \times 48$ for the pair, $n = (16 \times 240) / 48 = 3840 / 48 = 80$. Check: $\text{HCF}(80, 48) = 16$ and $\text{LCM}(80, 48) = 240$. The value 240 just repeats the LCM and 16 the HCF.

23. (A) For any pair of numbers the HCF must divide the LCM exactly. Check each: $240 / 16 = 15$, $280 / 14 = 20$, $450 / 15 = 30$ — all whole numbers, so those three are possible. But $250 / 18 = 13.9...$, not a whole number, so 18 cannot be the HCF when 250 is the LCM. Hence HCF 18 with LCM 250 is impossible.

24. (C) The largest equal number of bouquets is $\text{HCF}(480, 600, 720)$. Factorising: $480 = 2^5 \times 3 \times 5$, $600 = 2^3 \times 3 \times 5^2$, $720 = 2^4 \times 3^2 \times 5$. Lowest powers: $2^3 \times 3 \times 5 = 120$. So 120 bouquets. The value 240 does not divide 600, and 60 is not the largest.

25. (C) Each bouquet gets $600 / 120 = 5$ lilies (and $480/120 = 4$ roses, $720/120 = 6$ tulips). Picking the rose count 4 or the tulip count 6 by mistake is the trap.

26. (C) Convert to cm: 1134 cm by 756 cm. The largest square tile that divides both exactly has side HCF(1134, 756). $1134 = 2 \times 3^4 \times 7$, $756 = 2^2 \times 3^3 \times 7$; lowest powers give $2 \times 3^3 \times 7 = 378$ cm. The smaller values divide both but are not the largest.

27. (A) Tiles along the length: $1134 / 378 = 3$; along the width: $756 / 378 = 2$. Total = $3 \times 2 = 6$ tiles. Adding instead of multiplying, or using a smaller tile, gives the wrong counts.

28. (A) In centimetres the lengths are 800, 1400 and 2100. The greatest common piece length is HCF(800, 1400, 2100) = 100 cm. ($800 = 2^5 \times 5^2$, $1400 = 2^3 \times 5^2 \times 7$, $2100 = 2^2 \times 3 \times 5^2 \times 7$; lowest powers $2^2 \times 5^2 = 100$.) So 100 cm, i.e. 1 m, per piece.

29. (A) We need HCF(493, 551, 667). Each is a multiple of 29: $493 = 29 \times 17$, $551 = 29 \times 19$, $667 = 29 \times 23$. The cofactors 17, 19, 23 are distinct primes, so the HCF is exactly 29. The distractors are those cofactors, which divide only one number each.

30. (C) The capacity is HCF(221, 323, 391). Each is a multiple of 17: $221 = 17 \times 13$, $323 = 17 \times 19$, $391 = 17 \times 23$. Since 13, 19, 23 are distinct primes, the HCF is 17 L. The distractors are the cofactors that each divide only one drum.

31. (B) Plot side = HCF(1517, 1739) = 37 (since $1517 = 37 \times 41$ and $1739 = 37 \times 47$). Along the 1739 m side the number of plots is $1739 / 37 = 47$. Along the 1517 m side it would be 41, which is the trap option.

32. (D) Largest equal row size = HCF(1056, 792). $1056 = 2^5 \times 3 \times 11$, $792 = 2^3 \times 3^2 \times 11$; lowest powers $2^3 \times 3 \times 11 = 264$. Rows = $1056/264 + 792/264 = 4 + 3 = 7$. Using a half-size row of 132 doubles the count to 14.

33. (A) Bundle size = HCF(825, 675, 1125). $825 = 3 \times 5^2 \times 11$, $675 = 3^3 \times 5^2$, $1125 = 3^2 \times 5^3$; lowest powers $3 \times 5^2 = 75$. So 75 cards per bundle. The value 25 = 5^2 drops the common 3, and 150 does not divide 675.

34. (B) Bag size = HCF(144, 180, 240) = 12 kg ($144 = 2^4 \times 3^2$, $180 = 2^2 \times 3^2 \times 5$, $240 = 2^4 \times 3 \times 5$; lowest powers $2^2 \times 3 = 12$). Bags = $144/12 + 180/12 + 240/12 = 12 + 15 + 20 = 47$. The value 12 is just the bag size, 20 is only the pulses count, and 94 doubles the answer.

35. (C) In centimetres the lengths are 1105, 1547 and 1989. The rod length is HCF(1105, 1547, 1989) = 221 (since $1105 = 221 \times 5$, $1547 = 221 \times 7$, $1989 = 221 \times 9$). So 221 cm. The factors 13 and 17 (since $221 = 13 \times 17$) divide it but are not the greatest.

36. (D) They next coincide after LCM(24, 36, 48, 60). Highest powers: $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$, $60 = 2^2 \times 3 \times 5 \rightarrow 2^4 \times 3^2 \times 5 = 720$ minutes. 360 misses the

2^4 , and 1440 doubles needlessly.

37. (C) They next meet after $\text{LCM}(63, 84, 105) = 1260$ s. The fastest runner has the shortest lap time, 63 s, so completes $1260 / 63 = 20$ laps. Using the slowest (105 s) gives 12, and the middle (84 s) gives 15 — both traps for misreading 'fastest'.

38. (B) They next meet after $\text{LCM}(40, 48, 60) = 240$ s. The slowest cyclist has the longest lap time, 60 s, completing $240 / 60 = 4$ laps. The fastest (40 s) does 6 and the middle (48 s) does 5 — picking those misreads 'slowest'.

39. (D) They realign after the same number of teeth has passed both gears, which is $\text{LCM}(40, 48) = 240$ teeth. Since the meshing point advances by one tooth on each gear together, 240 teeth of the smaller gear pass (and 240 of the larger too). The product $40 \times 48 = 1920$ over-counts, and 8 is the HCF.

40. (D) They toll together every $\text{LCM}(6, 9, 12) = 36$ seconds. In 30 minutes = 1800 seconds there are $1800 / 36 = 50$ such coincidences after the start. Including the start would give 51; halving wrongly gives 25.

41. (B) All four coincide every $\text{LCM}(10, 15, 20, 25) = 300$ seconds. In 3600 seconds there are $3600 / 300 = 12$ intervals, giving 12 coincidences after the start plus the start itself = 13. Forgetting the start gives 12; using a wrong period of 60 s gives 61.

42. (B) Next coincidence after $\text{LCM}(14, 21, 35)$ minutes. $14 = 2 \times 7$, $21 = 3 \times 7$, $35 = 5 \times 7 \rightarrow 2 \times 3 \times 5 \times 7 = 210$ minutes = 3 h 30 min. Adding to 9:00 am gives 12:30 pm. Using 7 minutes (the HCF) gives 9:07; mis-adding gives 11:00 am.

43. (D) Next together after $\text{LCM}(42, 70, 84)$. $42 = 2 \times 3 \times 7$, $70 = 2 \times 5 \times 7$, $84 = 2^2 \times 3 \times 7 \rightarrow 2^2 \times 3 \times 5 \times 7 = 420$ s. 210 misses the 2^2 , and 14 is the HCF, not the LCM.

44. (C) They leave together every $\text{LCM}(45, 60) = 180$ minutes = 3 hours. Adding 3 hours to 6:00 am gives 9:00 am. Using the HCF 15 min, or LCM 90 min, gives the earlier wrong times.

45. (B) First reunion at $\text{LCM}(56, 70, 84) = 840$ s. A completes $840/56 = 15$ laps, C completes $840/84 = 10$ laps, so A does $15 - 10 = 5$ more. Using B (70 s $\rightarrow$ 12 laps) in place of one runner, or reading the totals as the difference, gives the wrong gaps.

46. (B) They coincide every $\text{LCM}(30, 45, 75)$ seconds. $30 = 2 \times 3 \times 5$, $45 = 3^2 \times 5$, $75 = 3 \times 5^2 \rightarrow 2 \times 3^2 \times 5^2 = 450$ seconds = 7 min 30 s. So at 10:07:30. Using 15 s (the HCF) or a wrong LCM gives the other times.

47. (D) Such a number is $\text{LCM}(12, 15, 20) \times k + 7$. $\text{LCM} = 60$, and the smallest value ($k = 1$) is $60 + 7 = 67$. Reporting just the LCM 60, or taking $k = 2$ (127), or subtracting instead of adding (53) are the traps.

48. (B) $\text{LCM}(8, 12, 18) = 72$. The greatest 3-digit multiple of 72 is $72 \times 13 = 936$; adding the remainder gives $936 + 5 = 941$. (Check $941 < 1000$ and $941 - 5 = 936 = 72 \times 13$.) Forgetting to add 5 gives 936; mis-taking the LCM gives the others.

49. (C) $\text{LCM}(18, 24, 30) = 360$. The number plus 13 must be a multiple of 360, so the smallest positive number is $360 - 13 = 347$. Check: $347 + 13 = 360$, divisible by all three. Adding 13 instead of subtracting (373), or stopping at 360, are the errors.

50. (A) When the remainder is the same but unknown, the divisor divides every pairwise DIFFERENCE. Differences: $5097 - 2961 = 2136$, $6873 - 5097 = 1776$, $6873 - 2961 = 3912$. $\text{HCF}(2136, 1776, 3912) = 24$. (24 divides all three differences; 48 does not divide 1776 evenly? $1776/48 = 37$ — it does, but 48 must also divide 2136: $2136/48 = 44.5$, so 48 fails.) Hence 24.

51. (B) Leaving remainder 1 on 2,3,4,5,6 means the number is $\text{LCM}(2,3,4,5,6) \times k + 1 = 60k + 1$. We also need $7 \mid (60k + 1)$. Test $k = 1 \rightarrow 61$ ($61/7$ not whole); $k = 2 \rightarrow 121$ (no); $k = 3 \rightarrow 181$ (no); $k = 4 \rightarrow 241$ (no); $k = 5 \rightarrow 301 = 7 \times 43$ (yes). So 301. The others fail the divisible-by-7 condition.

52. (D) In both cases the deficit is the same: $12 - 5 = 7$ and $18 - 11 = 7$. So the number is 7 less than a common multiple of 12 and 18: it equals $\text{LCM}(12, 18) \times k - 7 = 36k - 7$. The smallest positive value ($k = 1$) is $36 - 7 = 29$. Check: $29 = 12 \times 2 + 5$ and $29 = 18 \times 1 + 11$.

53. (A) Remainder 2 on 5, 6 and 8 means the number is $\text{LCM}(5, 6, 8) \times k + 2 = 120k + 2$. We also need 7 to divide $120k + 2$. Testing $k = 1, 2, 3, 4$ gives 122, 242, 362, 482 — none divisible by 7. At $k = 5$ the number is $602 = 7 \times 86$, which works. So 602 is the smallest. The trap 122 is the first candidate that fails the divisible-by-7 test.

54. (B) In each case the remainder is 7 less than the divisor ($16 - 9 = 7$, $24 - 17 = 7$, $40 - 33 = 7$). So the number is 7 short of a common multiple: $\text{LCM}(16, 24, 40) \times k - 7 = 240k - 7$. The smallest positive value is $240 - 7 = 233$. Check: $233 = 16 \times 14 + 9$, $233 = 24 \times 9 + 17$, $233 = 40 \times 5 + 33$.

55. (C) $\text{LCM}(8, 12, 18) = 72$. The smallest 4-digit multiple of 72: $72 \times 13 = 936$ (3-digit), $72 \times 14 = 1008$ (4-digit). So 1008. The number $1080 = 72 \times 15$ is a 4-digit multiple but not the smallest, and 1000, 1044 are not multiples of 72.

56. (A) Remainder 4 in each case means the number is $\text{LCM}(6, 9, 15) \times k + 4$. $\text{LCM} = 90$, so the smallest value above 4 is $90 + 4 = 94$. Reporting the LCM 90 alone, taking $k = 2$ (184), or mis-LCM-ing are the traps.

57. (D) Co-prime numbers above 1 with product 1591 must be its two prime factors: $1591 = 37 \times 43$, both prime. The pair 1 and 1591 violates 'each greater than 1'; $31 \times 51 = 1581$ and $41 \times 39 = 1599$, neither equals 1591.

58. (A) $9 = 3^2$, $16 = 2^4$, $35 = 5 \times 7$ share no prime between any pair, so they are pairwise co-prime. In 8, 12 share 2; in 6, 21 share 3; in 10, 14 share 2 — each of those sets has a pair with HCF > 1 .

59. (C) Any number divisible by both 18 and 24 is divisible by their LCM = 72 ($18 = 2 \times 3^2$, $24 = 2^3 \times 3 \rightarrow 2^3 \times 3^2 = 72$). It need NOT be divisible by 144 or 432, which require higher powers. 36 divides 72 but is not the largest guaranteed divisor.

60. (C) Common multiples of 16 and 18 are the multiples of $\text{LCM}(16, 18) = 144$. Multiples of 144 up to 600: 144, 288, 432, 576 — that is $\text{floor}(600/144) = 4$ of them. Using $16 + 18$ or a wrong LCM gives the larger counts.